

Power equations



1

a $x = 3$

b $x = 8$

c $x = \frac{8}{5}$

d $x = 6$

e $x = 2$

f $x = 6$

g $x = 5$

h $x = 2$

Exponential equations

2

a $x = 8$

b $x = \frac{2}{9}$

c $x = 3$

d $x = 0$

e $x = -2$

f $y = -3$

g $x = -3$

h $x = \frac{1}{3}$

i $n = \frac{1}{3}$

j $x = -2$

3

a i $2^{3x} = 2^2$
ii $x = \frac{2}{3}$

b i $2^{4x} = 2^{-1}$
ii $x = -\frac{1}{4}$

c i $2^{-10} = 2^{2x}$
ii $x = -5$

d i $2^{\frac{x}{2}} = 2$
ii $x = 2$

4

a $y = 4$

b $k = -2$

c $y = \frac{3}{2}$

d $x = 2$

e $x = 2$

f $x = \frac{7}{3}$

g $x = -7$

h $x = \frac{21}{19}$

i $y = \frac{11}{6}$

j $x = -\frac{41}{6}$

k $x = 5$

l $x = 1$

m $x = 1 + \frac{2k}{3}$

n $x = 5$

o $x = \frac{5}{2}$

p $x = 4, -1$

q $x = 3$

r $x = \frac{4}{1+n}$



Exponential equations and logarithms

5

a i $x = \frac{\log 5}{\log 13}$ ii 0.627

b i $x = \frac{\log \frac{1}{11}}{\log 5}$ ii -1.490

c i $x = \frac{\log 2}{\log 3}$ ii 0.631

d i $x = \frac{\log 6.4}{\log 4}$ ii 1.339

e i $x = \frac{\log 5}{\log 0.4}$ ii -1.756

f i $x = \frac{\log 3125}{\log 5}$ ii 5

g i $x = \frac{\log 6}{\log \frac{1}{2}}$ ii -2.585

h i $x = \frac{\log 19\,207}{\log 27}$
ii 2.993

6

- a i Step 2, when factoring the a out of the index it should multiply the logarithm and not add to it.
ii $a = \frac{\log 40}{\log 9}$
iii 1.679

- b i Step 3. The 2 and 89 are in the incorrect positions in the logarithm.
ii $a = \frac{\log 89}{\log 2}$
iii 6.476

7 $x = -2$

Logarithmic equations

8

a $y = 16\,807$

b $y = 4$

c $y = 2$

d $y = 8$

e $x = 36$

f $x = 1$



9

a $x = 4$

b $x = 2$

c $x = 81$

d $x = 100$

e $x = \frac{1}{1024}$

f $x = 2$

g $x = 8$

h $x = 5$

i $x = \frac{6556}{7}$

j $x = 15$

k $x = 6$

l $x = 32$

m $x = 137$

n $x = 4$

o $x = 2$

p $x = -10$

q $x = \frac{7}{5}$

10

a $x = 11$

b $x = 1, 6$

c $x = 16, 32$

d $x = 5$

e $x = 8$

f $x = \frac{343}{6}$

g $x = \frac{5}{16}$

h $x = 2 + \sqrt{11}$

i $x = 2$

j $x = \frac{81}{5}$

k $x = \frac{125}{4}$

l $x = 2$

m $x = -\frac{15}{7}$

n $x = 3$

o $x = 5$

11

a $y = -3$

b $y = \frac{1}{3}$

12

a $x = 2.15$

b $x = 0.13$

13

a $x = 9$

b $x = 0.05, 100$

c $x = 0.12$